

EXECUTIVE OVERVIEW · MAY 2026

AI-Powered Alcohol Label Verification

TTB Label Verification — Take-Home Implementation Overview
Vikram Siwach · May 2026

78%

Field Accuracy

Qwen v2 LoRA — 2× Claude Haiku zero-shot baseline

5

Days to Ship

End-to-end build, 91 commits, 6 inference adapters

7

Mandatory Checks

All 27 CFR mandatory fields implemented and verified



Manual Label Review at Federal Scale

TTB reviews approximately **150,000 alcohol label applications per year** with only 47 agents. A simple review takes 5–10 minutes — and nearly half of each agent's day is spent on mechanical matching: brand name, ABV, net contents, government warning. Prior automation attempts failed because they exceeded the non-negotiable 5-second latency threshold, and agents abandoned them.

150K

Label Applications

Submitted to TTB per year for review

47

Reviewing Agents

Covering the full intake today

5–10

Minutes Per Label

Current average for a simple review

Constraints from Stakeholder Interviews

≤ 5 Seconds

Non-negotiable latency ceiling — the deal-breaker for the prior vendor

Accessibility

Approachable for agents 50+ (half the workforce) and a 73-year-old benchmark user

Federal Firewall

Most outbound traffic to public ML endpoints is blocked at the network level

Batch Upload

200–300 labels at a time during peak-season importer submission flows

AI Reads. Rules Decide. Humans Own the Call.

The system separates perception, compliance logic, and human judgment into three distinct, auditable layers — designed so the legal logic never depends on any single model.



01 · Extraction

Vision-Language Model

Open-weights Qwen2.5-VL-7B (Apache 2.0), fine-tuned on TTB-stratified data.

Reads what's printed; reports observations and per-field confidence.

Never decides compliance.



02 · Verdict

Deterministic 27 CFR Engine

Hand-coded Python rules engine cites the regulation behind every match / likely / flag decision. 118 unit tests. Fully auditable, reproducible, and model-independent.



03 · Decision

Human-in-the-Loop

Agent confirms or overrides every result. Three-state output (match / likely / flag) gives a confidence signal without forcing a hard call. Engine routes conservatively.



The rules engine is the durable value — swap the model anytime, and the legal logic doesn't change.

Browser → Backend → Swappable Extractor → Rules Engine

Browser

React + Vite — responsive,
agent-friendly UI

Backend

FastAPI + Pydantic — structured
data contracts

Extractor

6 swappable adapters behind
one ABC interface

Rules Engine

Pure Python · 27 CFR · match /
likely / flag

Inference Modes — pick at runtime via
`INFERENCE_MODE`

mock

Deterministic fixtures · CI
pipeline

cloud

Anthropic / OpenAI /
Google APIs

claude-code

Routes via Claude Code
CLI

onprem

Tesseract + Phi-3.5-vision
(Ollama)

sft

Qwen v2 LoRA loaded locally on GPU

Three-Way Parallel Fan-Out

One Vercel function · three concurrent observation sources via `asyncio.gather` · deterministic rules engine · -5 s warm. Qwen handles PII fields on our boundary; Haiku handles public text only; Tesseract provides reproducible font-size measurement.

Modal Qwen v2 LoRA

4 PII fields · privacy boundary

- Brand name, Class & type, Bottler name/address, Country of origin
- Qwen2.5-VL-7B + LoRA r=16, BF16 · A100 GPU on a container we control
- Trained on N=8,968 TTB labels (3 epochs)
- Warm: -4 s · Cold start: 30–60 s

Anthropic Haiku 4.5

3 public-text fields · commodity

- Alcohol content, Net contents, Government Warning verbatim
- Tool-use structured extraction · no PII in payload
- Safe for commodity model — all 27 CFR 16.21 public text
- Warm: -1.5–3 s · Anthropic API

Modal Tesseract (CPU)

Deterministic 16.22 inputs

- Warning bbox height (px), Image DPI (EXIF), ABV / Net regex fallback
- Tesseract OCR, no GPU required
- Back-scaled to original-image px for reproducible `fontSizeMm` measurement
- Warm: -1–1.5 s

⚠ On Modal Qwen timeout (`MODAL_TIMEOUT=8s`, cold start): falls back to Haiku-only extraction + Tesseract overlay. `extractorSource='haiku-fallback'` is surfaced on the response for agent visibility.

Per-Field Source + 27 CFR Citation

Every field is routed to the most appropriate extractor based on data sensitivity and availability of agency-labeled training data. The rules engine cites the exact CFR regulation behind every match, likely, or flag decision — providing a fully auditable compliance trail.

Field	Extractor Source	Regulation Cited
Brand Name	Qwen v2 LoRA	27 CFR 4.32 / 5.63 / 7.63
Class & Type	Qwen v2 LoRA	27 CFR 4.32 / 5.63 / 7.63
Bottler Name & Address	Qwen v2 LoRA	27 CFR 4.35 / 5.65 / 7.65
Country of Origin	Qwen v2 LoRA	27 CFR 4.35 / 5.66 / 7.65
Alcohol Content	Anthropic Haiku	27 CFR 4.36 / 5.65 / 7.71
Net Contents	Anthropic Haiku	27 CFR 4.37 / 5.47 / 7.27
Government Warning	Tesseract OCR + Haiku verbatim	27 CFR 16.21 / 16.22

- ❏ Hybrid by design: the 4 fields with agency-clean training data run on-prem (Qwen v2 LoRA). The 3 without labeled data run on Haiku — documented as a known external dependency that disappears when agency data lands for those fields.

Qwen v2 LoRA: 77.5% on the 4 Trained Fields

Validated on a 2,000-row TTB-stratified holdout (1,028 wine / 689 spirits / 283 malt · seed 42). The result falls within noise of the original 200-row eval (78.0% → 77.5%), confirming the model generalizes — it is not overfit to the initial sample.

77.5%

Qwen v2 LoRA · micro field accuracy

2,000-row TTB-stratified holdout · fine-tuned on N=8,968
agency labels · 3 epochs

37.2%

Anthropic Haiku · zero-shot baseline

Same 2,000-row holdout · no fine-tuning · Haiku zero-shot held
at 37.2% on the 200-row eval as well

- ✓ The 2× improvement over Haiku zero-shot validates that fine-tuning on TTB-specific registry data — not just prompt engineering — is the path to production-grade accuracy on PII fields.

SECTION 6 · WHY V2 BEAT V1

Better Data + Narrower Schema → 77.5% on the 2K Holdout

V1 · COLA CLOUD BASELINE

Training Data

COLA Cloud free sample, N = 1,377

Schema

Wide — 4 fields + Government Warning text + image quality

Epochs

2

- ❌ 38.4% micro on COLA Cloud labels (in-distribution) · **1.9% on the 2K TTB-Registry holdout** — v1 never saw Registry images. The strong in-distribution score was entirely corpus-specific.

V2 · TTB REGISTRY · PRODUCTION TARGET

Training Data

TTB Public COLA Registry, stratified scrape, N = 8,968 (6.5× more)

Schema

Narrowed — 4 fields only · GW handled by deterministic OCR

Epochs

3

- ✅ 78.0% on the 200-row release eval · 77.5% on the 2,000-row holdout — within noise · zero hallucination · **88.5% on Class & type**

Two key takeaways: Volume matters — 6.5× more data plus 50% more epochs explains the bulk of the gain. Schema discipline matters — narrowing to fields with clean ground truth eliminated v1's warning-text hallucination entirely. The 2K out-of-distribution collapse from 38% → 1.9% proves v1's number was specific to its training corpus, not a general capability.

5 Days · 91 Commits · 6 Inference Adapters



Same Code · Privacy Boundary on PII

All three topologies run the identical verifier codebase and rules engine. The only variable is where the model inference executes — and that is controlled by a single environment variable.

1

A · Privacy-Focused Hybrid (*Currently Live*)

Vercel + Modal Qwen v2 + Haiku

React on Vercel CDN. Modal Qwen v2 LoRA on a container we control reads the 4 PII fields. Haiku reads 27 CFR 16.21 verbatim warning + ABV + net contents — all public-text values. PII never touches a third-party API.

Cost: \$0–5/mo + \$1/1,000 labels · **Privacy:** Modal boundary on PII · Haiku for 27 CFR public text

2

B · Privacy-Only (No Commodity Model)

Modal Qwen v2 only · Tesseract for warning

Same React + FastAPI. Modal Qwen v2 + Modal Tesseract container handle every field — Haiku is removed entirely. Government Warning verbatim falls back to Tesseract regex. Use when no third-party API is acceptable in the data path.

Cost: \$0/hr idle · \$0.001/req active · **Privacy:** Zero third-party API · single provider

3

C · Agency-Hosted (Production Target)

Azure Gov / Azure ML · on-prem GPU

Same Docker image inside FedRAMP boundary. `INFERENCE_MODE=sft` loads the LoRA on agency GPU. Zero outbound calls. Rules engine identical. Clears FedRAMP High, IL4/5 requirements.

Cost: Inside agency Azure spend · **Privacy:** FedRAMP High · IL4/5

❗ Swapping topology is one env var. The verifier code never changes.

Demo URL + Downloadable Weights for Independent Validation

Live Demo

ttb-label-verification-ebon.vercel.app

Mode: Modal Qwen v2 (PII fields) + Haiku (public text) hybrid

Latency: ~5 s warm · ~9 s cold-start fallback to Haiku

- **Samples:** 67 real TTB-approved labels (curated)
- **Checks:** All 7 mandatory fields + GW verbatim & casing

GitHub Release · v0.1.0 — Qwen v2 LoRA + Eval Pack

github.com/vsiwach/ttb-label-verification/releases/tag/v0.1.0

ttb_qwen_v2_adapter.zip

152 MB · 3-epoch, BF16,
N=8,968 — production model

ttb_qwen_v1_adapter.zip

146 MB · COLA Cloud baseline
(ablation reference)

ttb_eval_pack.zip

85 KB · 200-row v1/v2/Haiku predictions + scorer script

 Treasury can validate the on-prem path independently by downloading the v2 adapter and pointing any GPU host's deployment script at it — no vendor dependency required.

Decisions Made + What They Cost



Buy + Train, Not Buy or Train

Ran Claude (cloud) AND Qwen (open-weights LoRA) end-to-end on the same eval set in parallel. Refused to pick one before measuring. Result: v2 won decisively on trained fields; Haiku held on long-text reading. Both are in production — each where it wins.



Narrow the Schema

v1's wide schema (4 fields + warning text) caused hallucination on long verbatim text. v2 retrained fields-only; warning text handled by deterministic OCR + canonical-text matching in the rules engine. Hallucination on warning → 0.



Get Data from the Source

COLA Cloud's CCO mirror had OCR-spillage in brand/class fields. Scraped 20K labels directly from the TTB Public COLA Registry at 0.5 req/sec (polite). 6.5× the data was the headline change that pushed v2 to 78%.



Architecture Portability Is the Durable Value

Six interchangeable extractors behind one ABC interface. Treasury can swap Modal → Cerebrum → Azure ML → on-prem with one env var. The rules engine — the legal logic — never changes across providers.

From Prototype to Production

01

Move Inference Inside the FedRAMP Boundary

Deploy the same Docker image to Azure ML Online Endpoints inside Azure Government (FedRAMP High, IL4/5). Verifier code unchanged – `INFERENCE_MODE=azure_ml` is the only new piece. Sub-FedRAMP cloud APIs fully eliminated.

02

Expand On-Prem Coverage with Agency Data

ABV, Net contents, and Warning text currently use Haiku (cloud). When agency-labeled training data lands for those fields, the same LoRA pipeline absorbs them and Haiku exits the loop entirely. Privacy posture improves field-by-field.

03

Productionize the Deterministic Checks We Deferred

27 CFR 16.22 type-size (`font_mm` via Tesseract bbox + EXIF DPI) is wired but currently degrades when DPI is missing (~49% of labels). Capture DPI at COLA submission → check goes from "not assessable" to enforced.

Take-Home Repository

github.com/vsiwach/ttb-label-verification

Live Demo

ttb-label-verification-ebon.vercel.app